Outputs of Degradation Functions

Additional Test Cases

This appendix presents two representative test cases used to evaluate the robustness of the ResNet18 vision model under simulated underwater degradations.

Test Case 1: Turtle Image

Figure 1: Original turtle image

Figure 2: Turbidity simulation

Figure 3: Depth color shift simulation

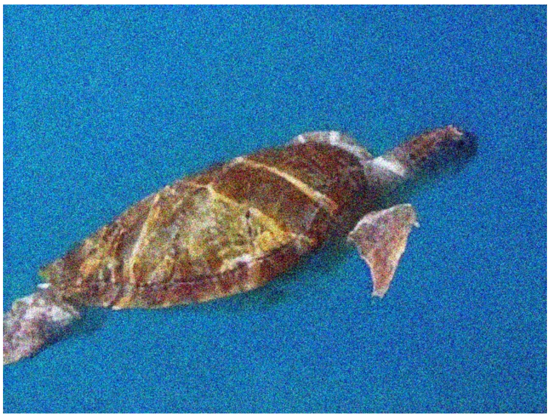

Figure 4: Sensor noise simulation

Condition	Prediction	Confidence	Latency (ms)
Baseline	loggerhead	79.38%	73.92
Turbidity	loggerhead	30.41%	54.80
Color Shift	loggerhead	58.57%	55.87
Sensor Noise	loggerhead	70.35%	54.47

The model consistently identified the turtle as **loggerhead** under all conditions. However, turbidity caused a significant confidence reduction, indicating that blur and haze strongly affect feature extraction.

Test Case 2: Aquarium Image

Figure 5: Original aquarium image

Figure 6: Turbidity simulation

Figure 7: Depth color shift simulation

Figure 8: Sensor noise simulation

Condition	Prediction	Confidence	Latency (ms)
Baseline	eel	48.04%	101.01
Turbidity	cardoon	4.96%	53.78
Color Shift	eel	57.95%	55.15
Sensor Noise	African chameleon	6.72%	48.42

This image demonstrates a more severe failure case. Under turbidity and sensor noise, the model produced incorrect predictions with extremely low confidence, highlighting the vulnerability of ImageNet-trained models to underwater degradations.

Summary

The turtle image demonstrates confidence degradation while preserving the correct class prediction. In contrast, the aquarium image shows complete prediction failure under certain degradations. Together, these examples illustrate the impact of environmental conditions on underwater perception systems.